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A000111(015)

B. Tech. PQ (First & Second Semester) Examination 2020

(AIC Scheme)

(AEI, Bio Tech, Chem., Civil, CSE, Elect., EEE, EI, Et & T, IT, Mech., Mining, Metallurgy, Mechatronics, Production, Automobile Agriculture, Plastic Branch)

PHYSICS-I

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt five units as per your branch. Part (a) of each question is compulsory and carries 4 marks. Attempt any two part from (b), (c) and (d) of each question carrying 8 marks. Do not use color ink (use only blue or black). Diagram should be neatly drawn and well labelled.*

1. (a) (i) Convert 10 kWh into mega joule. 1

(ii) How many par sec are there in one light year? 1

(iii) State the number of significant figure in $2{\cdot}67\times10^{-24}$ kg. 1

(iv) Calculate the angle of 1 degree and 1 min of arc in radian. 1

(b) Define trajectory of a projectile. Find the path of projectile, time of flight, maximum height and horizontal range when projectile is projected with velocity v making an angle θ with the horizontal direction. 8

(c) Define and derive expressions for position vector, velocity vector and acceleration vector with diagram. Discuss uniform circular motion and derive expression for the magnitude of centripetal acceleration. 8

(d) A particle is fired horizontally with a velocity of 100 m/s from the top of a hill 500 m high. Calculate (i) the time taken to reach the ground (ii) the distance of the target from the hill (iii) the velocity with which the projectile hits the ground. 8

2. (a) (i) Define moment of inertia of a rigid body. 1

(ii) Define bending stress and shear stress. 2

(iii) In rotational motion, what are the units of angular velocity and angular acceleration? 1

(b) State and prove the parallel axis theorem for moment of inertia of a rigid body. 8

(c) With the help of diagram derive relation between linear velocity and acceleration with angular velocity and acceleration in rigid body rotation about a fixed axis. 8

(d) (i) Calculate the angular momentum of disc whose rotational energy is 10 kJ and moment of inertia about the axis of rotation 8×10^{-4} kg-m^2. 4

(ii) A solid sphere of mass 50 g and radius 2 cm is rolling on a horizontal surface in a straight line with a velocity 5 cm/s. Calculate its total energy. 4

3. (a) (i) What pattern is formed at the center of Newton's rings? Is it dark or bright? Why? 2

(ii) State differences between interference and diffraction properties of light. 2

(b) Explain the formation of Newton's rings and show that the radius of dark rings is proportional to square root of natural numbers. 8

(c) Define Fresnel's Bi-prism and explain how this method is used to determine the thickness of a thin mica sheet. 8

(d) (i) In a Newton's rings experiment the diameter of the 15th ring was found to be 0·59 cm and that of the 5th ring is 0·336 cm. If the radius of curvature of the lens is 100 cm, find the wave length of the light. 4

(ii) A plane transmission grating has 16000 lines to an inch over a length of 5 inches. Find in the wavelength region of 6000 Å, in the second order (a) the resolving power of grating and (b) the smallest wavelength difference that can be resolved. 4

4. (a) Deduce the relation between electric field and electrostatic potential. 4

(b) Calculate of an electric field and electrostatic potential in a conducting spherical shell. 8

(c) What is dielectric polarization? Explain different types of dielectric polarizations. 8

(d) (i) Prove that : 4

$$\varepsilon_r = 1 + \chi$$

(ii) Show that the vector field

$$\vec{F} = \left(x^2 + xy^2\right) i + \left(y^2 + x^2 y\right) j$$

is irrotational. 4

5. (a) Explain Biot-Savart law and its importance. 4

(b) Give the detail classifications of magnetic materials. 8

(c) Calculate the magnetic field for a bar magnet in axial and equatorial position. 8

(d) (i) What are hard and soft magnetic materials? Give their applications. 4

(ii) A straight wire carries a current of 6A from south to north direction. What is the magnetic

field due to 1 cm piece of wire at a point 50 cm, 30 degree east of north. 4

6. (a) Explain Farady's laws of electromagnetic induction. 4

(b) What is Poynting vector? Explain Poynting vector in electromaghnetic field. 8

(c) Prove that : 8

(i) $\text{div}\,\vec{B} = 0$

(ii) $\text{div}\,\vec{D} = \rho$ and

(iii) $\text{curl}\,\vec{B} = -\dfrac{dB}{dt}$

(d) (i) Deduce the relation between electric and magnetic field of an electromagnetic wave. 4

(ii) Find the energy stored in a one meter length of a laser beam operating at 1 mW. 4

7. (a) Explain Heisenberg's uncertainty principle. On the basis of uncertainty principle, show the non existence of electron in the nucleus. 4

(Given : Planck constant = $6{\cdot}623\times10^{-34}$ j.s., size of nucleus = 10^{-14} m, speed of light = 3×10^{3} m/s)

(b) Explain De-Broglie's hypothesis of matter waves? Derive the expression for De-Broglie wavelength. How was the De-Broglie wavelength of electron determined and verified by Davisson and Germer Experiment? 8

(c) Derive Schrodinger's time dependent and time independent wave equation. 8

(d) (i) An electron has a speed of 600 m/s with an accuracy of 0.005%. Calculate the certainty with which we can locate the position of the electron.

(Given : Planck constant = $6{\cdot}62\times10^{-34}$ j.s., mass of electron = 9.1×10^{-31} kg) 4

(ii) An electron is trapped in a one dimensional infinite potential well of width 1Å. How much energy must be supplied to excite it from ground state to first excited state. 4

8. (a) Explain electron in a periodic potential. 4

Or

Explain Kronig-Penney model for energy bands in solids.

(b) Draw neat energy level diagram for conductyors, insulators and semiconductors and distinguish between them. 8

(c) Describe the formation of energy bands in solids and discuss the concept of direct and indirect band gap semiconductors. 8

(d) (i) The Fermi level for potassium is 2·1 eV. Calculate the velocity of the electrons at the Fermi level. 4

(ii) In a solid consider the energy level lying 0·01 eV below Fermi level. What is the probability of this level not being occupied by an electron? (kT = 0·026 eV, RT = 300 K). 4

9. (a) Explain the law of mass action for N-type and P type semiconductors. 4

(b) What do you mean by Fermi level in semiconductor? Show that Fermi level lies in midway of forbidden energy gap in intrinsic semiconductor. 8

(c) What is P-N junction? Define depletion region, barrier potential and describe its forward and reverse biased characteristics with energy band diagram. 8

(d) In a doped semiconductor, there are $4{\cdot}52\times10^{24}$ holes and $1{\cdot}25\times10^{14}$ electrons per cubic meter. What will be the carrier density in undoped specimen? Electrons and holes mobilities are 0·38 m^2/Vs and 0·18 m^2/vs respectively. Calculate the conductivity of intrinsic and the doped semiconductors. 8

10. (a) Explain : 4

(i) Total internal reflection

(ii) Stimulated emission of radiation

(b) Describe the construction and working of semiconductor laser. What are advantages of this laser. 8

(c) Discuss the structure of optical fibre and derive the expressions for acceptance angle and numerical aperture. 8

(d) A glass clad fibre is made with core glass of

refractive index 1.5 and the cladding is doped to give a fractional index difference of 0.0005.

Find :

(i) the cladding index

(ii) the critical internal reflection angle

(iii) the external critical acceptance angle and

(iv) the numerical aperture 8